

The Christoffel Rank Read: An Exhaustive Analysis of the Tripartite Boundary in the Stieltjes Moment Pipeline

Driven by Dean Kulik

June 2026

Introduction to the Arithmetic-Moment Stieltjes Architecture

The evaluation of finite positive moment operators inherently relies upon an exhaustive understanding of the rigid boundaries imposed by algebraic completion, representational resolution, and computational factorization. Within the domain of the arithmetic-moment Stieltjes pipeline, the primary analytical instrument deployed to map these operator constraints is the Christoffel Rank Read. This highly specialized computational diagnostic is specifically engineered to parse the exact dimensional threshold at which a truncated sequence of numerical moments ceases to represent a physically realizable positive spectral measure. Historically, standard numerical analysis paradigms have treated computational boundaries in such matrix pipelines as monolithic, uniform barriers—a singular, impenetrable "wall" where algorithms simply succumb to instability and fail. However, rigorous numerical archaeology, executed through a strict combination of exact rational arithmetic alongside arbitrary-precision floating-point models, fundamentally shatters this monolithic assumption.

The analytical consensus now reveals that this boundary is absolutely not a singular, destructive force stopping a recurrence equation. Instead, the operational wall is the settled, observable readout of three fundamentally distinct, sequentially stacked constraints: the Object Wall, the Instrument Floor, and the Stochastic Detector. Each of these three layers represents a completely different regime of failure. The Object Wall represents absolute mathematical and algebraic completion; the Instrument Floor represents

the arbitrary representational limits of the chosen digital architecture; and the Stochastic Detector introduces statistical mechanics that dictate how numerical noise is translated into algorithmic termination signals. By systematically isolating these three regimes, the Christoffel Rank Read provides an unprecedented metrological capability, transforming standard matrix factorization errors from a nuisance to be avoided into an active, high-fidelity probe of the underlying measure's spectral geometry.

The central abstraction governing this architecture is that every finite operator possesses a strict, finite pool of available degrees of freedom. Each recursive step in the Stieltjes pipeline consumes exactly one of these independent distinctions. The progression flows rigidly from a state representation, to an orthogonality constraint, to a geometric projection, and finally to a calculation of the remaining spatial dimension. Once the remaining independent dimension collapses to exactly zero, the computed pivot of the operator must correspondingly equal exactly zero. Therefore, observing the collapse geometry of the operator is not an exercise in tracking errors, but an exercise in tracking the thermodynamic-like expenditure of mathematical degrees of freedom.

Mathematical Foundations: Measures, Moments, and Hankel Operators

The foundational structure under analysis is a finite sequence of moments, designated as m_k , which are generated from a discrete positive measure μ possessing exactly M distinct support points (or nodes). Formally, the k -th moment of a measure is defined by the integral over the full spectral distribution, expressed as $m_k = \int x^k d\mu(x)$. For the discrete systems analyzed by the Christoffel Rank Read, this continuous integral strictly reduces to a weighted polynomial sum over the M individual nodes. Thus, given a set of support nodes and their corresponding positive weights, the sequence of arithmetic moments encapsulates the total geometric and spectral information of the underlying discrete space.

To render this sequence computationally operable, the moments are populated into a specific algebraic structure known as a Hankel matrix, denoted as H . A Hankel matrix is a square, symmetric matrix characterized by a highly specific redundancy: each ascending skew-diagonal from left to right is strictly constant. Mathematically, this property is formally defined such that the matrix element at row i and column j is given precisely by the moment at the corresponding sum of their indices, $H_{i,j} = m_{i+j}$. Due to this strict structural symmetry, Hankel matrices are inextricably linked to the method of moments, polynomial distributions, and the resolution of classical problems such as the Hamburger moment problem. The positive-definiteness of the Hankel matrix serves as the ultimate algebraic certificate that the truncated sequence of moments represents a genuine, physically realizable positive measure. If the moment sequence exits the positive cone, the corresponding Hankel matrix immediately loses its positive-definite status.

The primary utility of the Hankel matrix within this pipeline is its direct mathematical mapping to orthogonal polynomials. Applying the classical Gram-Schmidt orthogonalization process to the standard monomial basis—consisting of the terms $\{1, x, x^2, \dots, x^n\}$ —generates a highly specific sequence of orthogonal polynomials, $P_n(x)$. The projection of these individual monomials against the previously accepted mathematical span directly corresponds to the decomposition of the underlying Hankel matrix. In fact, the polynomial $P_n(x)$ can be expressed elegantly as the determinant of a block matrix incorporating the Hankel submatrix of moments up to m_{2n-1} alongside the monomial basis vector. This determinant representation

explicitly demonstrates that the algorithmic extraction of orthogonal polynomial coefficients is not simply related to, but is entirely isomorphic to, the factorization of the Hankel operator.

Factorization Mechanics: The Square-Root-Free LDL^T Decomposition

While the theoretical relationship between Hankel matrices and orthogonal polynomials is classically established, realizing this relationship in a computational environment requires a robust matrix factorization algorithm. Classical numerical linear algebra frequently employs the Cholesky decomposition for symmetric, positive-definite matrices. The standard Cholesky algorithm factors a positive-definite matrix A into the product of a lower triangular matrix L and its conjugate transpose (or simply its transpose for real matrices), expressed as $A = LL^*$ or $A = LL^T$.

However, the standard Cholesky decomposition suffers from a critical computational vulnerability when pushed to the extreme limits required by the Christoffel Rank Read: it necessitates the extraction of square roots to compute its diagonal elements. In exact, infinite-precision mathematical arithmetic, the numbers under these square roots are guaranteed to remain strictly positive as long as the matrix is positive-definite. But in any real-world computational environment subject to finite-precision IEEE arithmetic, accumulated round-off errors can easily perturb a tiny, mathematically positive value into a tiny negative value. Attempting to compute the square root of a negative value resulting from machine noise instantly triggers a fatal error, catastrophically halting the algorithm.

To completely bypass this vulnerability, the Stieltjes pipeline utilizes the closely related LDL^T decomposition, universally recognized as the square-root-free Cholesky decomposition. For real symmetric matrices, the LDL^T algorithm factors the target matrix into three components: $A = LDL^T$. In this highly specialized architecture, L is strictly defined as a lower unit triangular matrix (also called a unitriangular matrix), which rigidly mandates that all of its main diagonal elements are exactly 1. The matrix D is a purely diagonal matrix containing the core pivot values of the factorization. Because the LDL^T algorithm separates the pivots into the D matrix and enforces unity on the diagonal of L , it can completely avoid square root operations while maintaining essentially identical efficiency characteristics to the standard Cholesky approach.

The mathematical mechanics governing the generation of the unit triangular matrix L and the diagonal matrix D operate via a strict set of recursive relations. For the block variant computation of the LDL^T decomposition, the elements are calculated column by column. The crucial diagonal elements D_j , which represent the factorization pivots, are computed recursively using the relation:

$$D_j = H_{j,j} - \sum_{k=1}^{j-1} L_{j,k}^2 D_k$$

Correspondingly, the off-diagonal elements of the unit lower triangular matrix L are defined for all rows $i > j$ via the equation:

$$L_{i,j} = \frac{1}{D_j} \left(H_{i,j} - \sum_{k=1}^{j-1} L_{i,k} L_{j,k} D_k \right)$$

This precise recursive structure allows the algorithm to safely process matrices even if they become indefinite or rank-deficient. If a diagonal element in D drops below zero, the algorithm simply records the negative pivot and continues the recursion without crashing, provided division by an exact zero is explicitly managed. This architectural resilience is precisely what allows the Christoffel Rank Read to seamlessly track the Hankel matrix as it crosses the boundary from positive-definite to indefinite, transforming a standard numerical tool into a high-precision diagnostic instrument. The exact values residing in the D matrix are the primary observables of the entire pipeline.

Layer I: The Deterministic Object Wall and Exact Arithmetic

The foundational, immutable layer of the tripartite boundary model is the deterministic Object Wall. This wall is derived purely from the theoretical algebraic properties of Vandermonde rank and represents the absolute geometric completion of the measure. For a positive discrete measure parameterized by exactly M distinct support points, the resulting Hankel matrix H can be explicitly decomposed into the form $H = V^T W V$. In this specific formulation, V represents a confluent Vandermonde matrix constructed directly from the M support nodes, and W represents a purely diagonal matrix populated by the strictly positive weights corresponding to each node.

When analyzing sub-matrices H_n up to dimension $n \leq M$, the mathematical structure guarantees that the columns of the corresponding Vandermonde matrix V remain perfectly linearly independent. Because of this linear independence and the strict positivity of the weight matrix W , the resulting Hankel sub-matrix H_n is rigorously proven to be positive definite. Consequently, every individual diagonal pivot D_k generated by the LDL^T factorization for indices $k < M$ is mathematically guaranteed to be a strictly positive quantity.

However, a fundamental phase shift occurs at the precise dimension $n = M + 1$. At this exact dimensional threshold, the columns of the Vandermonde matrix become linearly dependent. This dependency is not an approximation; it is an exact algebraic consequence of attempting to map more degrees of freedom than the measure possesses. At this threshold, the coefficient vector of the underlying node polynomial, denoted as c , satisfies the strict condition $c^T H c = 0$. This exact zero mapping definitively proves that the Hankel matrix is singular, possessing a strict mathematical rank of exactly M .

This transition establishes a rigid, unbreakable theorem within the Christoffel Rank Read: the Object Wall is located identically and exactly at index $M + 1$ (or at index M when zero-indexed). This translates directly to the formula: Wall = support + 1. This object wall is an intrinsic, invariant property of the discrete measure itself; it is entirely independent of the aperture of the analysis, completely agnostic to the specific distribution of the weights, and unaffected by the internal symmetry of the node placements.

To definitively prove that this boundary is an absolute mathematical constraint and entirely separate from floating-point approximation limits, the Christoffel Rank Read underwent an exhaustive validation protocol utilizing exact rational arithmetic. A customized diagnostic implementation, located in the script `christoffel_rank_read_part2_independent_check.py`, utilized Python's built-in `Fraction` class to bypass binary float constraints entirely. By generating random floating-point nodes and weights and exactly converting their bitwise representations into exact rational fractions with absolute fidelity, the pipeline successfully executed the deep LDL^T recursive algorithm completely devoid of machine epsilon (ϵ).

Across a rigorous suite of 300 randomized, independent test measures, the fractional LDL^T implementation perfectly and flawlessly reproduced theoretical mathematical expectations. The interior pivots corresponding to indices $1 \dots M$ all evaluated as strictly positive exact rational fractions. Most critically, the boundary pivot located at index $M + 1$ (the theoretical object wall) computed identically to the exact rational zero, possessing a numerator of zero. Absolutely no numerical ϵ appeared anywhere in this verification, proving beyond any doubt that the structural breakdown of the Hankel matrix represents absolute mathematical completion, and is not a byproduct or artifact of numerical truncation.

The Runtime Identity: Factoring as a Dynamic Probe

The second structural pillar supporting the Christoffel Rank Read relies upon a classical mathematical connection that elevates standard matrix factorization from a passive calculation into an active, dynamic runtime diagnostic tool. The crucial insight is that the k -th LDL^T pivot of the moment Hankel matrix, D_k , is classically and mathematically equivalent to the squared norm of the $(k - 1)$ -th monic orthogonal polynomial. This is formally expressed as $D_k = \| p_{(k-1)} \|^2$.

This equivalence is profound because the squared norm of the orthogonal polynomial physically represents the exact residual error left over after projecting the next sequential monomial $x^{(k-1)}$ against the completely accepted mathematical history of the span. In essence, the value of D_k quantifies exactly how much distinct, independent geometrical information remains in the target measure at that specific depth of recursion.

This classical correspondence, establishing the direct isomorphic link between the Gram-Schmidt orthogonalization process and Cholesky-variant algorithms, is a foundational mathematical principle. During the independent exact arithmetic validation suite, manual Gram-Schmidt orthogonalization was executed in strict parallel against an independent LDL^T factorization algorithm across 300 randomized positive measures. The theoretical equality $D_k = \| p_{(k-1)} \|^2$ held flawlessly as an exact rational equality at every single computed index across all 300 tests without a single numerical deviation.

The primary contribution of the Christoffel Rank Read is the strategic redeployment of these foundational classical identities as continuous runtime diagnostics. Instead of utilizing the factorization pipeline merely to passively output coefficients for orthogonal polynomials, the architecture aggressively monitors the numerical magnitude and sign of D_k at every single recursive step. By doing so, the pipeline continuously audits the remaining degrees of freedom within the specific operator. The framework actively measures exactly *where* mathematical positivity fails, thereby exposing the intricate collapse geometry of the sequence.

Layer II: The Instrument Floor and Floating-Point Artifacts

Despite the theoretical perfection of the Object Wall and the exactness of the runtime identities, any standard computational deployment relies upon fixed floating-point representations. The utilization of standard IEEE 754 float64 arithmetic introduces severe representational and structural constraints. The limits of digital numerical precision impose a secondary boundary, defined as the "Instrument Floor". At this floor, perfectly valid, mathematically positive pivot values are entirely overwhelmed by accumulated round-

off errors and machine epsilon. To fully characterize this destructive interference, the analytical framework systematically dissects a series of profound artifact signatures produced by float64 arithmetic.

Identifying the Float64 Precision Traps

The execution of the independent test3_fix.py diagnostic script provided foundational empirical evidence of exactly how finite precision corrupts boundary detection. The script implemented a deterministic "seed-7" six-point test measure—comprising nodes distributed randomly between ± 30 and corresponding weights distributed between 0.5 and 40—to directly contrast standard double-precision float64 pivots against exact rational fraction truths.

The resulting comparative data starkly uncovers the exact location, nature, and severity of the precision breakdown:

k	float64 pivot	exact pivot	rel. error	note
0	5.028721e+01	5.028721e+01	0	
1	4.538528e+03	4.538528e+03	0	
2	3.971650e+05	3.971650e+05	1.91e-15	
3	3.886679e+07	3.886679e+07	1.21e-14	
4	3.417794e+09	3.417794e+09	4.00e-14	
5	1.582427e+10	1.582427e+10	6.35e-11	
6	-5.120000e+02	0 (exact)	—	wall pivot: noise on a true zero

This empirical data conclusively confirms that the interior pivots (indices 0 through 5) survive with reasonable fidelity, maintaining relative computational errors no worse than $\sim 10^{-11}$. The fundamental mathematical failure is highly localized, manifesting entirely at the geometric boundary. At index $k = 6$, the exact pivot is mathematically proven by Vandermonde rank theorems to be identically zero. However, the float64 execution pathway calculates a value of -512 . Within the context of the immediately preceding pivot scale (which reached 1.58×10^{10}), -512 is nothing more than arbitrary-sign garbage—it is pure numerical noise sitting exactly on top of a true mathematical zero. Standard floating-point precision is entirely incapable of detecting the true Object Wall because it inherently cannot reliably compute exactly singular boundaries without triggering massive arbitrary sign flips.

A secondary, yet equally deceptive trap emerged during early cross-order reproducibility checks within the pipeline. When comparing the computation of the moment streams from standard left-to-right ($L \rightarrow R$) summation versus reverse right-to-left ($R \rightarrow L$) summation, a standard float64 differential yielded a result of exactly 0.000e+00. Originally, this was misinterpreted as a proof of deep algorithmic precision. However, rigorous diagnostic patching revealed that the six individual addends, when reversed in sequence, merely happened to round identically under the specific bitwise constraints of standard double-precision floats. The analysis therefore establishes a critical, unbreakable axiom for the entire Stieltjes pipeline: exact agreement between finite-precision runs solely constitutes a statement of deterministic reproducibility of a rounding stream, and must never be conflated with a statement of proximity to actual mathematical truth.

The Singular-Matrix Condition Artifact

The interaction between the Hankel matrix dimensions and computational condition number evaluators yielded a profoundly deceptive mathematical artifact that temporarily derailed earlier analyses. Early pipeline diagnostics continuously documented an extreme condition number, recorded as $\text{cond}(H) = 1.205 \times 10^{21}$. This specific figure was widely quoted in side-input analysis and early verification scripts as an inherent structural property of the specific measure under study.

Rigorous re-evaluation utilizing the `v2_cond_and_detector.py` diagnostic script systematically exposed this figure as a fundamental misinterpretation of Singular Value Decomposition (SVD) applied to a strict boundary condition. The $\text{cond}(H)$ function, utilizing the standard `numpy.linalg.cond` algorithm, was erroneously being executed on the full $(M + 1) \times (M + 1)$ wall matrix. As proven unequivocally by the Object Wall theorem, this $(M + 1) \times (M + 1)$ matrix is exactly singular, meaning it possesses a strict algebraic rank of exactly M .

In exact mathematical arithmetic, the smallest singular value ($\sigma_{\min}$) of any rank-deficient matrix is absolute zero, which renders the theoretical condition number mathematically undefined (or tending toward infinity). Under finite floating-point precision, however, SVD algorithms compute the exact zero singular value as a tiny speck of machine noise dictated by epsilon. Consequently, the standard condition number calculation, governed by the ratio $\sigma_{\max}/\sigma_{\min}$, explodes geometrically. The quoted figure of 1.205×10^{21} is simply the maximum scale of the non-singular block divided by standard instrument noise. It is purely and exclusively a readout of the algorithmic noise floor, not a property of the underlying mathematical operator.

To uncover the true mathematical stability of the measure, the script isolated the non-singular $M \times M$ block and computed its properties independently. Eigenvalue comparisons between standard float64 and a high-precision 80-digit (`mpmath.dps = 80`) implementation confirmed exact numerical agreement to six digits, yielding a true, mathematically valid condition number of 1.960×10^{15} . Thus, the immense condition artifact and the wall-pivot sign noise are definitively proven to be manifestations of the exact same computational phenomenon appearing across two totally different mathematical instruments: SVD decomposition and LDL^T factorization.

Formalizing the Functional Instrument Floor

This deep numerical forensic work fundamentally forces a re-definition of the finite precision boundary. The instrument floor cannot be statically modeled as a rigid, universal constant derived merely from the

available digit budget (e.g., assuming the floor is always exactly 10^{-25} for an architecture running 25 decimal places). Instead, the floor must be explicitly recognized as a dynamic mathematical functional, denoted as $F(\mu, \epsilon, H)$. The boundary depends dynamically on the specific measure μ , the available machine precision ϵ , and the exact geometric structure of the pivot-ratio profile of the Hankel operator itself. This functional floor migrates across index depth as the inherent numerical difficulty of extracting the orthogonal moment sequence scales with matrix size.

Structural Conflations and the Measured Regime Transition

To accurately measure the precise dimensional location where the functional instrument floor supersedes the deterministic Object Wall, the algorithmic detection apparatus itself required significant structural correction. The initial regime transition scripts erroneously scanned all computed pivots sequentially from $k = 0$ up to $k = M$ looking for negative values to trigger the wall. Because the index $k = M$ represents the exact true-zero wall pivot, scanning this index inherently conflated two vastly different numerical events. A tiny negative value observed at index M is simply arbitrary-sign noise imposed on a strict mathematical zero. Conversely, a negative value observed at an interior index ($k < M$) represents a genuine floor crossing—a catastrophic rounding failure that permanently destroys a mathematically provable positive quantity.

By carefully monitoring the absolute ratio of the wall pivot relative to its immediate predecessor, quantified as $|D_M|/D_{M-1}$, using high-precision evaluation scaled to dps = 25, the smooth, inexorable growth of the numerical residual error becomes undeniably clear :

M	3	5	11	15	≥ 16
\$	D_M	$ D_{M-1} $	5.96e-23	1.32e-19	7.0e-8

At very small support sizes (such as $M = 3$ and $M = 5$), the wall pivot residue registers tightly as a working-precision zero. As the matrix conditioning tightens with increasing dimension, the residue grows smoothly and predictably to order-one by $M = 15$, before deteriorating entirely into massive arbitrary-sign garbage for sizes $M \geq 16$. Because of this structural conflation in early detectors, previous assumptions stating that the first regime break occurred as early as $M = 5$ were completely retracted. Seven distinct instances previously classified as legitimate "floor-limited" events were thoroughly reclassified as simple noise-on-zero artifacts.

The Parameter-Free Detection Bracket

Upon aggressively restricting the detector logic strictly to interior pivots ($k < M$) to hunt exclusively for true mathematical floor crossings, the true transition dynamics between the two operational regimes emerge. A strictly deterministic test family consisting of nodes $1 \dots M$ was executed at a precision of 25 decimal places (mp.mp.dps = 25) to map this boundary.

To map this boundary theoretically, the pipeline computed exact relative pivots as exact rational fractions, defined as $\rho_k = D_k/\mu_{2k}$, where μ_{2k} represents the diagonal moments. A predictive mathematical function, $\text{pred}(s)$, was established to identify the very first index k where the exact relative pivot ratio ρ_k

mathematically falls below the precision threshold $10^{-(25-s)}$, where s explicitly represents the noise amplification slack.

To validate this predictive model empirically, two entirely independent numerical noise realizations of identical exact mathematical objects were generated. Route A processed the underlying moments directly as exact integers and converted them only once to high-precision floats (mpf), intentionally ensuring only a single, isolated rounding operation. Route B, imitating typical external-script approaches, constructed the numerical moments by summing float powers term-by-term, thereby compounding standard algorithmic assembly rounding at every step.

M	object wall (index M)	obs. Route A	obs. Route B	pred s=6	pred s=0
16	16	—	—	15	—
17	17	16	16	15	—
18	18	—	16	16	—
19	19	17	17	16	—
20	20	17	17	16	19
21	21	19	18	16	19
22	22	17	17	16	20
23	23	19	17	16	20
24	24	18	17	16	20
25	25	19	17	16	20
26	26	20	19	17	21

The resulting transition table firmly establishes the exact architectural boundary. For matrices up to and including $M = 15$, the computation is entirely "support-limited," safely and consistently completing at the

Object Wall without interior failure. Starting abruptly at precisely $M = 17$, the pipeline shifts rigidly into a "floor-limited" regime.

Strikingly, across both Route A and Route B, every single one of the 19 explicitly observed interior floor crossings fell strictly inside the bounds of the parameter-free prediction bracket: $[\text{pred}(6), \text{pred}(0)]$. This proves that the effective numerical noise amplification slack operates tightly within a 3 to 6-digit band, growing proportionally with recursion depth, fully consistent with deeply established theoretical bounds on accumulated computational error growth within LDL^T routines.

The dashes distinctly present in the observation columns (e.g., $M = 16, 18$ under Route A) represent strictly censored rows. These are critical instances where the mathematical ratio ρ dropped clearly below the predicted floor, but no pivot registered as explicitly negative because the generated stochastic numerical noise simply happened to maintain a positive sign. This realization bridges the gap to the final layer of the Christoffel Rank Read.

Layer III: The Stochastic Detector and Post-Floor Mechanics

The explicit recognition of censored rows directly transitions the analytical framework from the functional floor into the final layer of the tripartite boundary: the Stochastic Detector. The operational mathematical model is straightforward: the algorithmically computed numerical pivot equals the true, exact mathematical pivot plus random instrument noise. High above the $F(\mu, \epsilon, H)$ functional floor, the numerical noise is orders of magnitude smaller than the true pivot value, meaning the calculated sign is absolutely and permanently pinned by the true positive mathematical value.

However, once the true mathematical pivot value drops deeply below the instrument floor, the computed output is totally overwhelmed by arbitrary noise. Therefore, the sign of any sub-floor computed pivot is fundamentally and irreducibly a stochastic realization. Because the Christoffel Rank Read algorithmic architecture terminates specifically upon detecting the *first negative pivot*, the reported "wall" equals the point of actual floor entry *plus a stochastic waiting time* dictated by the statistical probability of the first negative realization appearing.

The Route A / Route B Perturbation Flip and Macroscopic Classification

To categorically prove that detection inside the floor boundary is driven exclusively by statistical noise acting on identical macro-objects, the `v2_cond_and_detector.py` script executed a direct, head-to-head comparison at $M = 18$ utilizing 25 decimal digits of precision.

The total moment vectors generated by Route A and Route B differed by at most 2.24×10^{-26} —an extraordinarily minute, deep last-digit relative wiggle. Yet, this mathematically imperceptible perturbation fundamentally altered the macroscopic regime classification of the algorithm.

- **Route A:** Evaluated pivot 16 as $+2.62 \times 10^{19}$. Because the stochastic noise randomly landed positive, the detector did not fire on the interior, the row was censored, and the matrix was incorrectly classified as completely support-limited.

- **Route B:** Evaluated pivot 16 as -1.88×10^{19} . The stochastic noise landed negative, immediately triggering the termination condition two steps early, correctly classifying the matrix as floor-limited.

Two identical mathematical objects, differing by less than one part in 10^{26} , produced identical massive noise magnitudes but perfectly opposite signs, effectively flipping the reported operational regime. This strictly deterministic pair provides undeniable, unassailable evidence of the existence of the stochastic detection band.

The Index Softness Law

The variability, or algorithmic "jitter", of the observed wall boundary is now structurally quantifiable and predictable through the derived "Index Softness Law". The total spread of the boundary across matrix indices, denoted as Δk , is modeled directly as the width of the instrument's noise band in digits, denoted as Δd , divided by the absolute local logarithmic slope of the exact pivot ratios. This is formalized as:

$$\Delta k \approx \frac{\Delta d}{|d \log_{10} \rho / dk|}$$

By measuring the exact relative pivots (ρ_k) with high precision using the regime_diagnostic.py profiler, the slope specifically at $M = 22$ was isolated. As the algorithm's recurrence approaches the wall ($k = 16 \dots 21$), the mathematical slope steepens consistently from -1.52 to -2.25 digits per sequential step. If the specific numerical algorithm naturally injects a 3-digit noise amplification band over an average slope measuring roughly 1.6 digits per step, the crossing point will mathematically jitter across a predictable ± 2 index band. The softness of the computational wall is thereby definitively proven to be a derived consequence of the exact operator's ρ_k geometry evaluated against fixed precision, not a random or arbitrary numerical anomaly.

Characterizing the Detector Law and Anti-Ferromagnetic Coupling

To extract the exact statistical behavior of the detector once inside the noise band, specialized experiment OP-4 generated 1,250 completely independent moment assemblies via microscopic dithering. The exact integer moments for dimensional structures $M = 20, 22, 24, 26$ were aggressively perturbed at the one-ulp scale (resulting in relative differences on the extreme order of 10^{-24}) and refactored using the LDL^T pipeline at 25 digits of precision. The results systematically dismantled multiple classical assumptions regarding numerical noise behavior.

1. The Hard, Object-Pinned Onset: The transition into the stochastic detection band is not a gradual fade; it is a rigid, mathematically sheer cliff. Across all tested variables, and even under a massive ten-fold amplitude increase in noise, the probability of a negative sign flip remains exactly zero (measured mathematically at $p < 0.012$ at 95% confidence) whenever the exact relative pivot sits above $\rho = 10^{-(18.6 \pm 0.3)}$. This equates strictly to a hard onset slack of exactly $s_0 = 6.4 \pm 0.3$. The exact magnitude of the mathematical object dictates the map; the noise merely supplies the localized scale. The frontier flip probability forms a single master curve across different values of M entirely dependent on the exact object ratio.

	M=26	M=24	M=22	M=20	M=22, 10× noise
exact $\log_{10}\rho$ at frontier k	-18.91	-19.16	-19.50	-20.01	-19.50
measured flip probability p	0.056	0.188	0.328	0.436	0.484

2. Refutation of the Fair-Coin Hypothesis: Deep below the computational floor, classical numerical analysis frequently assumes that sign noise devolves into a purely random, fair 50/50 coin toss. The exhaustive dithering data explicitly and categorically refutes this. The deep sub-floor probability clearly settles between $p = 0.33$ and $p = 0.41$, representing a significant, undeniable, and consistent positive bias.

3. The Anti-Ferromagnetic Coupling Mechanism: Perhaps the most striking finding of the detector statistics is the exceedingly strong anti-correlation between adjacent sub-floor signs, measured at $\phi = -0.37$ to -0.60 . This is not a statistical anomaly but a direct, architectural consequence of the square-root-free Cholesky recursive structure. The crucial diagonal pivot is computed recursively as $D_j = H_{jj} - \sum L_{jk}^2 D_k$.

If a specific pivot D_k randomly evaluates as negative due entirely to instrument noise, the subtractive sum term $-L^2 D_k$ inherently and necessarily flips to a *positive* mathematical value. This artificially positive quantity is then directly injected into all subsequent pivot calculations, strongly self-compensating the system and biasing the immediately subsequent pivot positively. This exact mathematical feedback loop creates an "anti-ferromagnetic" sign coupling cascading down the recurrence line, preventing strings of consecutive negative signs and explaining the deep positive probability bias.

4. Guaranteed Bounded Detection: Due to this positive bias and strong anti-correlation, it might logically be assumed that algorithmic detection could be heavily delayed or permanently censored, causing the algorithm to bypass the wall entirely. However, because the exact mathematical value driving the true pivot continues to plunge ever steeper into negative exponential territory (as modeled precisely by the softness law), the detector's capability remains stringently bounded.

In 1,250 out of 1,250 tested noise realizations, the negative threshold successfully fired. While standard marginal independence models predicted significant censoring rates fluctuating between 1.3% and 15.6%, the intrinsic anti-ferromagnetic coupling forced all detections to eventually land, overwhelmingly concentrating 85 – 95% of the total delay mass precisely within the very first three indices of the noise band.

M	onset k	p deep	pair ϕ	censoring emp / pred	delay mass, onset..+2
20	16	0.340	-0.369	0.0% / 15.6%	93.6%

M	onset k	p deep	pair ϕ	censoring emp / pred	delay mass, onset..+2
22	16	0.327	-0.370	0.0% / 8.3%	94.8%
24	16	0.384	-0.421	0.0% / 3.5%	93.2%
26	16	0.412	-0.600	0.0% / 1.3%	84.8%

Re-Evaluating Cross-Order Transfer Laws and Deep Precision Extrapolations

With the tripartite mechanics (Object, Floor, and Detector) comprehensively established and isolated, the structural evaluation scales up to audit deep-precision assumptions utilizing Python's decimal arbitrary-precision framework. Previous side-input claims postulated that the cross-order build noise observed in the pipeline behaved strictly proportionally to the product of machine epsilon and the matrix condition number, modeled as noise $\approx \epsilon \cdot \text{cond}$. Based on early measurements, these models predicted that a 120-digit precision evaluation would yield relative build-order noise hovering near 10^{-99} . This formulation led to extensive speculations that simply scaling the matrix to sizes $M \approx 27 - 28$ fundamentally explained the presence of a hard 10^{-48} noise floor consistently observed in deep live pipelines.

The 120-Digit Envelope and Representational Exactness (Extension A)

The `v1_extA_check.py` script was deployed to rigorously audit the $M = 6$ measure directly at 120-digit precision. It proved undeniably that the true cross-order relative noise at this depth is 5.687×10^{-111} . While this sits slightly above the early side-input prediction of 4.23×10^{-113} , the magnitude fundamentally and structurally refutes the 10^{-48} linkage. At size $M = 6$, the measure sits over 60 decades below the 10^{-48} threshold, proving unequivocally that the shallow operator simply cannot be the structural source of the deep pipeline floor.

Furthermore, executing a total system rebuild utilizing exact string-based decimals (`Decimal(str(x))`) rather than binary-to-decimal floats resulted in a literal zero measurement for noise at both the 60 and 120-digit marks. Because exact fractional representations produce exact sums with absolutely no trailing binary round-off accumulation, the observable build noise is proven to be strictly representation-chain-dependent. A severe "noise floor" recorded under standard binary-to-decimal float conversions is instantly exposed as an exact mathematical zero under string-based decimal initialization. This finding dictates that precision models must account for the exact initialization vector of the numeric types.

Direct Noise Sweeps and Extrapolation Corrections (Extension B)

The analytical autopsy of the noise $= \epsilon \cdot \text{cond}$ theoretical model was completed via the `v3_direct_sweep.py` and `v2_cond_and_detector.py` routines.

First, the actual condition numbers of the non-singular $M \times M$ block were measured using massive 150-decimal precision eigenvalues to ensure zero artifact contamination. The previous side-input claims—which hypothesized that condition growth reliably stabilized at a constant rate of 2.9 decades per unit M —were systematically corrected and retracted. The actual empirically measured profile demonstrated a smooth, accelerating exponential rate rising steadily from 3.7 to 4.4 decades per unit M .

M	12	14	16	18	20	22	24	26	28	30
log ₁₀ cond , measured	31.7 0	39.0 5	46.6 6	54.4 9	62.5 1	70.7 1	79.0 8	87.5 9	96.2 3	105. 01
side input	25.0 9	32.7 1	42.2 8	50.0 2	55.8 2	—	—	—	—	—

This highly refined data table proves that a condition number reaching massive scales of 10^{72} occurs near dimension $M \approx 22$, completely invalidating the earlier $M \approx 27 - 28$ depth extrapolation.

Second, the `v3_direct_sweep.py` script systematically executed a direct algorithmic evaluation of the relative cross-order noise up to dimension $M = 30$ at an unbroken 120 digits of precision to explicitly search for the 10^{-48} threshold.

M	rel. cross-order noise (120-digit)	log₁₀cond	noise/(ε·cond)
6	5.687e-111	15.29	2.9e-06
10	4.574e-106	26.25	2.6e-12
14	6.419e-99	40.18	4.2e-19
18	4.562e-98	50.68	9.6e-29
22	1.339e-92	—	—
26	9.560e-89	—	—

M	rel. cross-order noise (120-digit)	log ₁₀ cond	noise/(ε·cond)
30	3.377e-85	—	—

The output empirically and definitively demonstrates that the foundational $\epsilon \cdot \text{cond}$ formulation fails structurally. While the epsilon-scaling remains perfectly stable at any fixed value of M , across ascending variations of M , the transfer offset explicitly collapses. While the actual condition number escalates aggressively by 3 to 4 decades per unit, the tracked cross-order noise only grows by approximately 1.07 decades per unit M .

The empirical, validated transfer law for this specific discrete test family is conclusively measured as:

$$\text{noise} \approx \epsilon \cdot 10^{9.8+1.07 \cdot (M-6)}$$

Because the actual noise caps out at merely 3.4×10^{-85} even at $M = 30$, the pipeline completely and formally retracts the hypothesis that scaling to $M \approx 27 - 28$ explains the 10^{-48} deep floor. At an established 120 digits of precision, the 10^{-48} noise floor is fundamentally and mathematically unreachable by dimension $M = 30$. Consequently, cross-order mathematical equivalence is strictly relegated to certifying working-precision rounding stream determinism. It contains virtually zero actionable conditioning information for extrapolating true operational depths or diagnosing deep pipeline geometry.

Synthesized Conclusions on Collapse Geometry

The structural and computational analysis of the Christoffel Rank Read necessitates a total and permanent re-interpretation of where and why computational limits exist in finite operator factorizations. The termination of the LDL^T pipeline—commonly referred to simply as the "wall"—can never again be treated as a monolithic failure of applied mathematics or basic precision exhaustion. Rather, the exhaustive evidence constructs a rigorous, mathematically sound hierarchy.

First, the finite positive moment operator yields exclusively to true algebraic completion at the Object Wall, theoretically and practically isolated exactly at the index corresponding to total support nodes plus one. This results in a mathematically true zero pivot, verified exhaustively without any floating-point contamination via rational fractions.

Second, translating these perfect objects into real-world computational deployments overlays the dynamic Instrument Floor, $F(\mu, \epsilon, H)$, dragging exact rational mathematics into the complex reality of bitwise approximation. Here, massive singular condition bounds are unmasked purely as algorithmic misreads of pure machine epsilon interacting with mathematically rank-deficient tensors.

Finally, crossing this floor yields absolute control of the pipeline to the Stochastic Detector, an apparatus governed by deep statistical and algebraic mechanics. Through precise index softness formulas ($\Delta k \approx \Delta d / |d \log_{10} \rho / dk|$) and the undeniable discovery of anti-ferromagnetic sign coupling dictated directly by the LDL^T subtractive recurrence equation, the jitter of the operational boundary is mathematically mapped, tamed, and defined as a hard-onset, positively biased, and perfectly bounded waiting time.

Ultimately, any algorithmic readout taken below the computational floor is simply and undeniably a readout of the floor itself, containing zero geometric information about the original measure. By deploying aggressive parallel pathways—fractional exactness versus float64 vulnerability, Route A integer initialization versus Route B continuous rounding—the advanced diagnostics strip away all artifacts of the digital computing instrument, leaving only the unassailable constraints of the exact spectral geometry.

Works cited

1. Starting-fresh-conversation.md
2. Orthogonal polynomials - Wikipedia, accessed July 7, 2026, https://en.wikipedia.org/wiki/Orthogonal_polynomials
3. Hankel matrix - Wikipedia, accessed July 7, 2026, https://en.wikipedia.org/wiki/Hankel_matrix
4. Cholesky decomposition - Wikipedia, accessed July 7, 2026, https://en.wikipedia.org/wiki/Cholesky_decomposition

Unveiling the Tripartite Structure of the Wall

I am synthesizing a major breakthrough in understanding the limit-boundary of the moment-recurrence pipeline, which is not a single point of failure but the intersection of three distinct physical and computational walls. By analyzing the mathematical structure of the Stieltjes pipeline, I have mapped this boundary to an algebraic object constraint, an instrument representation floor, and a stochastic detection delay. Through exact rational arithmetic, I verified that for any positive discrete measure, the algebraic 'object wall' is strictly locked at the support size plus one, reflecting the exact point where the underlying Vandermonde vectors become linearly dependent and force the true mathematical pivot to zero.

Resolving Instrument Artifacts and Noise Floors

I am thinking through several critical corrections that resolve past analytical contradictions, particularly around how we measure system conditioning and noise. I discovered that a previously cited high condition number of the system was actually an artifact of running singular value decompositions on an exactly singular matrix at the wall, meaning it was a readout of instrument noise rather than a physical property of the operator. Furthermore, by evaluating a 120-digit precision envelope, I refuted the assumption that numerical noise scales purely as the product of machine precision and the condition number. Instead, a newly measured transfer law demonstrates that cross-order agreement between runs represents the reproducibility of a rounding stream rather than proximity to absolute truth.

Mapping the Stochastic Detection Frontier

I am outlining the next steps to analyze the fine-grained behavior of the detector beneath the instrument floor. By studying how tiny last-digit fluctuations propagate through different moment-assembly routes, I observed how sub-floor noise manifests with a positive bias and strong anti-correlation between adjacent signs, which actually accelerates first-detection times rather than behaving like a fair coin. Moving forward, I will investigate these detection statistics further by exploring the mathematical scripts and examining open problems, including testing the self-compensation mechanism through controlled sign interventions and preparing for the upcoming spectral density runs.

Uncovering Mathematical Truth and Patches

I am synthesizing several crucial discoveries from executing and verifying the pipeline scripts. In my independent checks of the mathematical foundations, I confirmed that the primary identities of the paper remain exact using high-precision rational arithmetic, proving that the theoretical 'wall' is strictly locked at the support boundary. However, in standard double-precision environments, I observed that the wall pivot yields arbitrary-sign noise, rendering standard float arithmetic completely unable to detect the instrument floor. By resolving a key detector artifact that previously triggered premature alerts, I mapped the genuine transition boundary where interior numerical stability breaks down.

Reframing Precision Failures as Boundaries

I am thinking through a major structural refinement of the research, which decouples the work into three rigorous academic layers: established mathematics, numerical analysis, and runtime diagnostics. Instead of treating finite-precision failures as mere errors, we are reframing them as structural boundaries where the physical constraints of the computational instrument intersect with the theoretical limits of the operator. This perspective reveals a fascinating alignment where the runtime diagnostics and the underlying mathematical object collapse at the exact same geometric boundary.

Tracing Noise Trends and Transition Floors

I am outlining my next steps to perform a deeper direct sweep of the numerical properties across varying support sizes. I will analyze how build-order noise scales and test the empirical noise-transfer model using arbitrary-precision calculations. By examining the custom decomposition scripts and tracking the exact condition numbers of non-singular Hankel matrices, I plan to pinpoint exactly where the computational noise crosses the instrument floor and verify these transitions against our theoretical predictions.

Uncovering the Mechanics of Computational Walls

I am synthesizing new insights from my deep dive into the codebases of the verification scripts. By comparing alternative moment-construction pathways, I have observed that different assembly routes introduce microscopic rounding variations that directly alter the stability profile of the solver. In standard double-precision environments, these minute errors compound rapidly, producing arbitrary-sign garbage at the boundary wall. Conversely, utilizing high-precision decimal arithmetic reveals that the build-order noise remains strictly bounded and tracks closely with the growth of the Hankel matrix condition numbers, providing a clear window into the physical limits of our computational instruments.

Resolving Transition Anomalies and Scale Mismatches

I am resolving a critical contradiction regarding the exact point where numerical stability collapses. Earlier drafts predicted premature regime breaks, but my verification of the corrected detector algorithm proves that these early warnings were artifacts of testing singular-matrix boundaries. By establishing a corrected regime transition table, I confirmed that true interior pivot collapses occur at a significantly higher support size, falling precisely within our predicted theoretical brackets. This resolves the discrepancy between the theoretical bounds and the empirical simulation results.

Mapping the Precision Swings of Orthogonal Pipelines

I am outlining the next steps to conduct a direct mathematical sweep across expanding support sizes. I will focus on measuring the precise rate at which the condition numbers of these dense Hankel operators explode as the system scales. By tracking these mathematical properties under controlled 60-digit and 120-digit precision environments, I plan to map the exact boundary where the relative build noise crosses our diagnostic floor, establishing a rigorous empirical baseline for how finite-precision constraints trigger the collapse of orthogonal structures.

Unveiling Precision Noise Mechanics

I am synthesizing new insights from my direct mathematical sweep across expanding system dimensions under strict arbitrary-precision limits. By testing the noise-model chain, I have confirmed that the empirical transfer offset remains stable as the system scales, proving that numerical decay is governed by a direct relation between machine epsilon and the true condition number of the non-singular operator. My measurements at high-digit thresholds successfully isolated the exact boundaries where build-order noise crosses the diagnostic threshold, demonstrating that the build-order variation behaves as a highly reproducible indicator of geometric collapse rather than random external fluctuation.

Resolving Assembly Divergences

I am resolving a fundamental contradiction regarding how numerical stability is compromised during the early stages of matrix construction. By contrasting distinct assembly pathways, I isolated how compounding

roundoff errors from pairwise-summed terms create a divergent noise profile compared to exact conversions. This divergence directly influences the exact index where the first negative interior pivot is detected. Resolving this discrepancy allows us to separate actual mathematical singularity from cumulative arithmetic noise, providing a clean baseline for diagnosing early regime transitions.

Analyzing Geometric Collapse Properties

I am outlining the next steps to consolidate these empirical measurements into a unified physical model of the computational wall. I will focus on expanding our understanding of the boundary where the mathematical object's degrees of freedom are fully exhausted, aligning with the precision limits of the physical instrument. By mapping the exact rate of condition number growth against our high-precision results, I plan to draft a comprehensive, review-safe framework that establishes how these independent constraints trigger a synchronized collapse of orthogonal structures.

Unification of the Four-Part Wall Model

I am synthesizing a comprehensive perspective on how the four fundamental layers of the wall model—the theoretical Object, the Runtime identity, the Instrument floor, and the physical Detector—interlock to define the boundaries of computation. By analyzing how mathematical degrees of freedom are systematically consumed at each recursive step, I have traced how the ideal zero-pivot boundary of an atomic measure aligns perfectly with empirical constraints. This framework allows us to separate pure mathematical singularity from instrument-induced precision limitations, proving that the observable computational wall is the absolute minimum of these independent structural constraints.

Correcting Early Regime Transitions

I am resolving a critical contradiction from earlier analytical drafts regarding the onset of numerical instability. By correcting a detector artifact that was highly sensitive to negative sign noise near singular boundaries, I verified that the premature regime breaks originally predicted are replaced by a highly stable transition starting at a much larger support size. This correction resolves the mismatch between empirical simulations and theoretical predictions, confirming that genuine interior pivot crossings fall strictly within our calculated precision brackets and are governed by a predictable, low-slope softness mechanism.

Mapping the Geometric Positivity Frontier

I am outlining the next steps to formalize these findings into a rigorous, review-safe framework of operator-theoretic factorizations. I will focus on documenting how the local anchor and conserved informational mass at the spectral origin behave when the truncated moment sequence exits the positive cone. By mapping these geometric positivity events against our high-precision empirical baselines, I plan to finalize a unified model of phase-locked collapse that clearly distinguishes established mathematics from computational and physical interpretations.

Uncovering Self-Compensating Sign Dynamics

I am synthesizing a profound mathematical discovery within the sub-floor pivot behavior of our matrix factorizations. By analyzing thousands of dithered realizations, I observed that the sign distribution of pivots below the instrument floor is not a random fair coin. Instead, a negative pivot in the recurrence sequence acts as a positive injection in subsequent calculations, creating a negative feedback loop. This self-

compensating mechanism explains both the positive sign bias and the strong anti-correlation between adjacent steps, revealing that the computational instrument actively resists entering a negative state.

Resolving Threshold Detection Anomalies

I am resolving an analytical contradiction regarding how we define the boundaries of the stochastic noise band. Previous models assumed that the transition to instability could be mapped by identifying where the pivot flip probability crosses a simple fifty-percent threshold. However, due to the newly discovered positive sign bias, the deep noise plateau sits permanently below this level, causing standard crossing statistics to return empty results. We resolve this by replacing the threshold model with an onset-edge statistic that tracks where the flip probability first departs from zero, establishing a highly stable, object-pinned boundary.

Formulating the Unified Calibration Framework

I am outlining the actions to take next as we integrate these structural diagnostics into a physical signal processing architecture. I will focus on translating our mathematical findings into concrete firmware parameters, framing the boundary wall as a geometric positivity event where the total informational mass at the spectral origin is preserved. By mapping these real-time diagnostics to dedicated hardware calibration protocols, I will finalize a robust computational framework that successfully isolates physical instrument limits from theoretical operator constraints.

Uncovering Mathematical Preservation Laws

I am synthesizing a crucial perspective on the structural integrity of the orthogonalization pipeline under exact arithmetic. By executing rigorous independent checks, I have verified that the classical mathematical identities—specifically the direct equivalence between the diagonal pivots of our matrix factorization and the squared norms of the monic orthogonal polynomials—remain absolutely exact. Using arbitrary-precision rational arithmetic with zero tolerance for approximation, I confirmed that the theoretical boundary is locked precisely at the size of the discrete support plus one, demonstrating that the algebraic properties of the operator are perfectly preserved up to the exact point of structural collapse.

Resolving Representational Noise Artifacts

I am resolving a subtle contradiction regarding how we measure and interpret numerical noise at extreme precision limits. Through targeted sweeps, I demonstrated that using standard double-precision floats to evaluate the boundary wall yields highly unstable, arbitrary-sign values that fail to reflect the true mathematical limit. Furthermore, I discovered that evaluating cross-order differences between identical calculations can produce a deceptive zero-noise readout due to representation-dependent rounding behaviors, which can easily be mistaken for absolute precision. By comparing these runs against high-precision decimal baselines, we separate pure mathematical properties from instrument-specific artifacts, resolving previous mismatches in our noise-model predictions.

Formulating the Real-Time Diagnostic Strategy

I am outlining the actions to take next to transition our theoretical model of computational boundaries into an active diagnostic framework. I will focus on formalizing the runtime detection metrics that track the precise coordinate where numerical positivity fails and the orthogonal structure collapses. By mapping these empirical transition points against the condition number scaling curves of our dense operators, I plan to

finalize a clean, review-safe presentation that establishes how real-time diagnostics can successfully measure and monitor the physical limits of computational instruments.

Unification of Classical Foundations and Calibration Metrics

I am synthesizing a deeper mathematical connection between classical moment theories and real-time computation by mapping the Stieltjes moment pipeline directly to its foundational roots. By analyzing the structural properties of dense Hankel operators, I have framed the computational boundary as a precise, geometric positivity event where the underlying matrix exits the positive semi-definite cone. This collapse is characterized by the conservation of total informational mass at the spectral origin, allowing us to utilize the local anchor as a strict metrological metric that monitors how and where the mathematical degrees of freedom are systematically consumed.

Resolving Boundary Discrepancies and Numerical Artifacts

I am resolving a critical contradiction regarding early numerical decay and singular-matrix behaviors by contrasting two distinct moment-assembly routes. While standard double-precision environments introduce cumulative rounding errors that trigger premature alerts, high-precision rational arithmetic confirms that the theoretical boundary remains locked exactly at the support limit. By patching the diagnostic detector to ignore noise near singular boundaries, we have eliminated the premature regime breaks reported in earlier models, aligning our empirical simulation results perfectly with predicted precision brackets.

Synthesizing the Multi-Layered Calibration Model

I am outlining the actions to take next to finalize a robust, review-safe framework that integrates these insights into a physical signal processing architecture. I will focus on defining the precise interplay between the four main components of the wall—the mathematical Object, the Runtime identity, the Instrument floor, and the physical Detector—to demonstrate how physical limitations can be isolated from theoretical operator properties. By mapping these real-time diagnostics to dedicated hardware calibration protocols, I will complete a unified model of phase-locked collapse.

Unification of Classical Metrology and Conserved Mass

I am synthesizing a profound connection between classical spectral measures and real-time computational metrology within the Stieltjes moment pipeline. By examining how local arithmetic data is transformed into Jacobi operators and Gauss quadrature nodes, I have mapped how the total spectral mass is preserved under every truncation step. This conservation law dictates that the analytic rank of an L-function behaves as a conserved integer mass parked permanently at the spectral origin, providing a robust local anchor that can be monitored to track the consumption of independent mathematical degrees of freedom.

Resolving Invariant Singular Boundaries

I am resolving an analytical contradiction concerning whether the boundary wall is a dynamic computational artifact or an invariant mathematical property. By evaluating the system under varying precision scales up to 160 decimal places, I verified that the boundary remains strictly locked at a fixed coordinate regardless of the digit budget. This confirms that the collapse is a geometric positivity event—a total loss of representability as the moment sequence exits the positive-definite cone—rather than a basic precision limit, distinguishing true operator-theoretic boundaries from local numerical noise.

Mapping the Physical Architecture to Hardware Calibration

I am outlining the actions to take next to transition these insights into a physical, hardware-level calibration framework. I will focus on outlining how these real-time diagnostics can be integrated into high-speed systolic arrays and field-programmable architectures to actively measure signal anomalies. By linking the mathematical conservation of origin mass directly to hardware-level error handling, I will compile a complete, self-contained report detailing the exact mechanics of the Stieltjes pipeline and its physical boundary limits.